EXPERIMENT-13

01/10/25

IMPLEMENT YOUR OWN PING PROGRAM

AIM:

To implement a simple Ping program in computer networks to check whether a destination host is reachable & to measure the round-trip time (RTT) of packets.

PROCEDURE:

→ Start the program.

→ Read the destination host name or IP address from the user.

→ Create a ICMP packet.

→ Construct an ICMP echo Request packet containing:

- → ICMP type
- → code
- → Checksum
- → Identifier
- → Sequence Number
- → Data

→ Calculate the checksum of the packet.

→ Send the ICMP Echo Request to the destination.

→ Wait for an ICMP Echo Reply.

→ Calculate the round-trip time between sending the request & receiving the reply.

→ Display the destination, packet-size, sequence number, & RTT.

→ Repeat the process for a specified number of packets.

→ Display the final result & terminate.

PROGRAM:

```
import socket
import struct
import time
import os

def checksum(data):
    total = 0
    length = len(data)
    for i in range(0, length-1, 2):
        total += (data[i] << 8) + data[i+1]
    if length % 2:
        total += data[-1] << 8
    total = (total >> 16) + (total & 0xffff)
    total += total >> 16
    return (~total) & 0xffff

def ping(host):
    try:
        dest_addr = socket.gethostname(host)
    except socket.gaierror:
        print("Unable to resolve host")
        return
    icmp = socket.getprotobyname("icmp")
    socket.settimeout(2)
    packet_id = os.getpaid() & 0xFFFF
    sequence = 1
    header = struct.pack("!BBHHH", 8, 0, 0, packet_id,
                         sequence)
    data = b"Computer Science"
    packet = header + data
    chksum = checksum(packet)
    start_time = time.time()
```

```
try:
    sock.sendto(packet,(dest_addr,1))
    sock.recvfrom(1024)

    end_time = time.time()
    rtt = (end_time - start_time) * 1000

    print("Reply from", dest_addr, ":", "bytes =", len(packet),
          "time = %.02f ms" %rtt)

    except socket.timeout:
        print("Request timed out")

    finally:
        sock.close()

#Main Program
host = input("Enter hostname or IP address:")

for i in range(4):
    ping(host)
    time.sleep(1)
```

RESULT:

Thus, a ping program was successfully implemented using Python. The program sends ICMP Echo Request packets to the specified host, receives Echo Reply packets and calculates the round-trip time (RTT) to determine network connectivity.

SAMPLE OUTPUT:

Enter hostname or IP address : 8.8.8.8

Reply from 8.8.8.8 : bytes = 43

time = 24.51 ms

Reply from 8.8.8.8 : bytes = 43

time = 23.87 ms

Reply from 8.8.8.8 : bytes = 43

time = 24.36 ms

